THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 1 & 2: **A Square and a Cube, Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of 11^3 is: 1

(a) 1331 **(b)** 363 **(c)** 1342 **(d)** 1320

2. $\sqrt[3]{1728}$ is: 1

(a) 12 **(b)** 14 **(c)** 11 **(d)** 13

3. Which of the following is a perfect cube? 1

(a) 600 **(b)** 700 **(c)** 512 **(d)** 500

4. The square of an even number is always: 1

(a) odd **(b)** prime **(c)** even **(d)** a multiple of 3

5. The smallest number by which 54 must be multiplied to get a perfect cube is: 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 2

6. The number 4500 in standard (scientific) form is: 1

(a) 0.45×10^3 **(b)** 4.5×10^4 **(c)** 45×10^3 **(d)** 4.5×10^3

7. The number 0.00008 in standard (scientific) form is: 1

(a) 8×10^{-5} **(b)** 8×10^{-4} **(c)** 8×10^5 **(d)** 0.8×10^{-5}

8. One lakh is equal to: 1

(a) 10^7 **(b)** 10^6 **(c)** 10^5 **(d)** 10^4

9. 2×10^{-3} written in usual form is: 1

(a) 0.0002 **(b)** 2000 **(c)** 0.002 **(d)** 0.02

10. $(2^3)^0$ equals: 1

(a) 8 **(b)** 0 **(c)** 6 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the rule for the number of non-square numbers between n^2 and $(n+1)^2$. 2

12. Write the square of 105 using the pattern for numbers ending in 5. 2

13. Find x if $5^x = 625$. 2

14. Compare 3×10^4 and 2×10^5. 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Compare the growth of squares and cubes: which is larger for 12, 12^2 or 12^3? By how much? 3

16. If $3^{x+1} = 81$, find x and verify. 3

17. Why does a perfect square have an odd number of factors? Illustrate with 36. 3

OR

Find the smallest square number divisible by 10, 15 and 20. 3

18. Arrange in ascending order: 3×10^{-3}, 2×10^{-2}, 5×10^{-4}. 3

OR

A sheet of paper is 0.1 mm thick. Find the height of a stack of 10^4 sheets in metres. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Find the cube root of -1728. 5

(ii) Find $\sqrt[3]{0.027}$ and $\sqrt[3]{\dfrac{27}{125}}$.

OR

(i) State and verify the identity $(a+b)^2 - (a-b)^2 = 4ab$ using $a = 9,\ b = 4$. 5

(ii) Hence find $13^2 - 5^2$ using $(a+b)(a-b)$.

20. (i) Show that $(ab)^n = a^n b^n$ for $a = 2,\ b = 3,\ n = 3$. 5

(ii) Hence find $12^3 \div 6^3$.

OR

(i) Why do scientists prefer scientific notation? 5

(ii) Write the mass of an electron, 0.00000000000000000000000000000091 kg, in scientific notation.

END OF QUESTION PAPER — Total: 40 Marks

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